ONLINE LEARNING PLATFORM ANALYTICS USING SQL

Project Objective

To analyze purchase data to understand **sales trends**, **learner behavior**, and **popular course categories**.

Problem statement

To perform the below analysis.

- Display each learner's total spending (quantity × unit_price) along with their country.
- Find the top 3 most purchased courses based on total quantity sold.
- Show each course category's total revenue and the number of unique learners who purchased from that category.
- List all learners who have purchased courses from more than one category.
- Identify courses that have not been purchased at all.

Table Information

Learners Table

Field_Name	Data Type	Description
learner_id	INT	Unique id for learners
full_name	VARCHAR(100)	Learners name
country	VARCHAR(100)	Learners country

Course Table

Field_Name	Data Type	Description
course_id	INT	Unique id for course
course_name	VARCHAR(100)	Name of course
category	VARCHAR(100)	Course category
unit_price	DECIMAL(10)	Price per course

Purchase Table

Field_Name	Data Type	Description
purchase_id	INT	Unique id for purchase
learner_id	INT	Learners id
course_id	INT	Course id
quantity	INT	Quantity of purchase
Purchase_date	DATE	Date of purchase

Insights (SQL Analysis)

Data Exploration Using Joins -- Format currency values to 2 decimal places.Use aliases for column names (e.g., AS total_revenue). Sort results appropriately (e.g., highest total_spent first).

Purchase_ID	Learners_Name	Learners_Country	Course_Name	Total_Revenue
1012	Rekha	USA	JavaScript Essentials	14,994.00
1009	Narmada	USA	Python for Data Analysis	10,497.00
1002	Amudha	India	Advanced Python Programming	9,998.00
1017	Padma	India	Advanced Python Programming	9,998.00
1013	Elakiya	USA	Power BI Fundamentals	9,995.00
1001	Aditya	Singapore	Intermediate Python	8,997.00

INNER JOIN

Combine learner, course, and purchase data.

purchase_id	Learners_Name	Learners_Country	course_name	category	unit_price	quantity	purchase_date
1000	Abinaya	India	Python for Beginners	Python Development	1,499.00	2	2026-01-03
1018	Suchithra	India	Python for Beginners	Python Development	1,499.00	4	2026-04-05
1001	Aditya	Singapore	Intermediate Python	Python Development	2,999.00	3	2026-02-01
1011	Ravi	Singapore	Intermediate Python	Python Development	2,999.00	2	2026-02-20
1016	Narmada	USA	Intermediate Python	Python Development	2,999.00	3	2026-04-05
1002	Amudha	India	Advanced Python Programming	Python Development	4,999.00	2	2026-01-16

Display each learner's purchase details (course name, category, quantity, total amount, and purchase date).

learner_id	full_name	country	course_name	category	quantity	total_amount	purchase_date
108	Gayathri	Singapore	SQL for Beginners	SQL & Databases	3	3,897.00	2026-04-05
109	Elakiya	USA	Power BI Fundamentals	Data Visualization	5	9,995.00	2026-04-01
110	Karthika	Singapore	Excel for Data Analysis	Data Analysis	3	4,497.00	2026-03-04
111	Kavitha	USA	NULL	NULL	NULL	NULL	NULL
112	Lavanya	Singapore	Advanced SQL & Optimization	SQL & Databases	1	4,499.00	2026-04-05
113	Magizhini	India	NULL	NULL	NULL	NULL	NULL

Insights (SQL Analysis)

Q1. Display each learner's total spending (quantity × unit_price) along with their country.

learner_id	full_name	country	Total_Spending
100	Abinaya	India	2,998.00
125	Suchithra	India	5,996.00
101	Aditya	Singapore	8,997.00
122	Ravi	Singapore	5,998.00
115	Narmada	USA	19,494.00
102	Amudha	India	9,998.00

Q2. Find the top 3 most purchased courses based on total quantity sold.

course_name	Total_quantity_sold
JavaScript Essentials	9
Intermediate Python	8
Python for Beginners	6

Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.

category	Total_Revenue	Number_of_unique_learners
Cloud Computing	1,799.00	1
Data Analysis	14,994.00	2
Data Visualization	9,995.00	1
Machine Learning	11,997.00	2
Python Development	52,982.00	7
SQL & Databases	8,396.00	2

- Python development is the course category has the highest revenue and has been purchased by more learners.

Q4. List all learners who have purchased courses from more than one category.

learner_id	learners_name
108	Gayathri
115	Narmada

- Gayathri and Narmada are the 2 learners who purchased course from more than one category.

Q5. Identify courses that have not been purchased at all.

course_id	course_name
14	Tableau for Data Analysis

- Tableau for Data Analysis was the only course that was not purchased by any learners.

Q6. Top 5 courses based on total Revenue

course_name	Total_Revenue
Intermediate Python	23,992.00
JavaScript Essentials	22,491.00
Advanced Python Programming	19,996.00
Python for Beginners	16,489.00
Machine Learning Basics	11,997.00

- Intermediate python is the online course that has generated the highest Revenue.

Q7. Course on high demand based on quantity sold

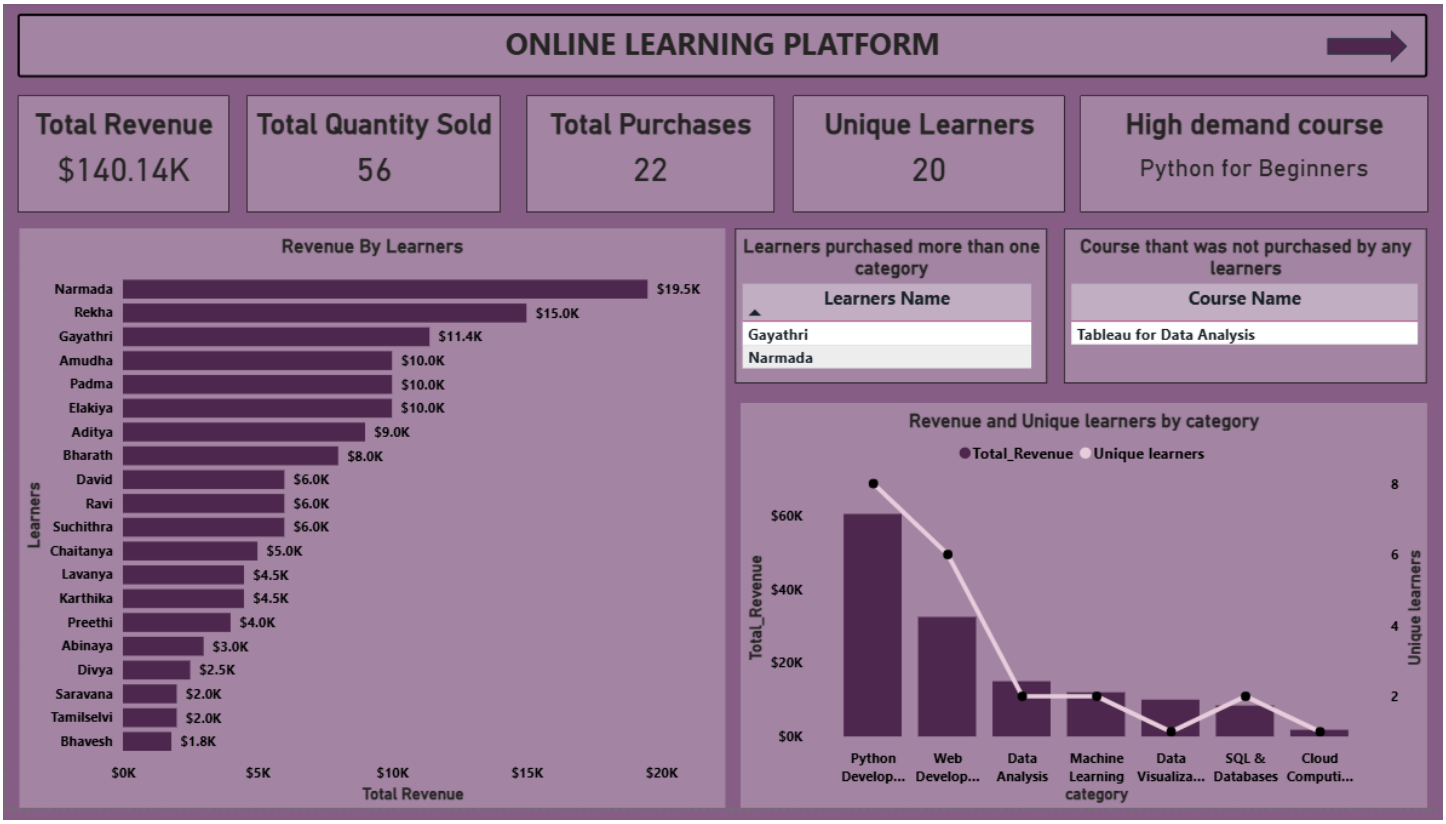
course_name	Total_quantity_sold
Python for Beginners	11

- Python for Beginners is the course on high demand.

Q8. Purchases by Country

country	Total_purchases	Total_Revenue
USA	6	48,280.00
Singapore	7	41,382.00
India	5	33,988.00
Canada	3	14,496.00
UK	1	1,998.00

Visualization Using Power BI



Key Insights

- Python category generates the highest revenue and demand
- Intermediate Python → highest revenue-generating course
- Python for Beginners → most popular (highest quantity sold)
- Most learners purchase from only one category
- Very few learners explore multiple categories
- Some courses have zero purchases (low visibility/demand)
- Revenue and engagement vary across different countries
- Significant difference in performance across course categories

Recommendations

- Focus on high-performing categories (Python, Data-related courses)
- Introduce advanced and specialized courses
- Improve cross-selling (course bundles & recommendations)
- Promote low-performing courses (discounts, marketing, content improvement)
- Create structured learning paths (Beginner → Advanced)
- Run country-specific marketing campaigns
- Use popular courses to drive up-selling opportunities

Conclusion

- Platform growth is driven by Python courses
- Strong beginner-to-intermediate learning trend observed
- Opportunity to improve cross-category engagement
- Need to optimize underperforming courses
- Strategic focus can boost revenue, retention, and scalability